

IDTA 02097

Safety Instrumented System (SIS) for the Process Industries

September 2026

SPECIFICATION

Submodel Template of the
Asset Administration Shell



Submodel Template

IDTA approved

- 100% AAS compliant
- Consistent & interoperable
- Released by the AAS experts

IDTA 02097

Imprint

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Chapter 1. General

1.1. About this document

This document is a part of a specification series. Each part specifies the contents of a Submodel template for the Asset Administration Shell (AAS). The AAS is described in [1], [2], [3] and [6]. First exemplary Submodel contents were described in [4], while the actual format of this document was derived by the "Administration Shell in Practice" [5]. The format aims to be very concise, giving only minimal necessary information for applying a Submodel template, while leaving deeper descriptions and specification of concepts, structures and mapping to the respective documents [1] to [6]. Common terms and abbreviations can be found in [7] to [11].

The target audience of the specification are developers and editors of technical documentation and manufacturer information, which are describing assets in smart manufacturing by means of the Asset Administration Shell (AAS) and therefore need to create a Submodel instance with a hierarchy of SubmodelElements. In addition to these people working at vendors or Engineering companies, it is interesting for people at owner operators working within the functional safety lifecycle. This document especially details on the question, which SubmodelElements with which semantic identification shall be used for this purpose.

This SMT will only be fully supported as of metamodel V3.0.

1.2. Scope of the Submodel

Safety is commonly defined as the absence of unacceptable risk, and risk reduction is achieved by implementing safety barriers that prevent accidents. Functional safety ensures that instrumented barriers operate correctly when needed, making it a critical element in accident prevention. IEC 61511 is an international standard that governs the use of Safety Instrumented Systems (SIS) in the process industry, such as chemical, pharmaceutical, and energy sectors. It outlines requirements for managing the entire safety lifecycle—from initial risk assessment and system design to operation, maintenance, and eventual decommissioning. A key concept in the standard is the use of Safety Integrity Levels (SILs), which define the required performance of safety functions to reduce risks to acceptable levels. The standard also emphasizes the importance of a structured management system, including clear responsibilities, documentation, and periodic reviews to ensure ongoing safety. Based on the more general IEC 61508, IEC 61511 adapts its principles specifically for the needs and challenges of the process industry.

There is an increasing number of digitalization initiatives related to SIS, motivated by the opportunity to reduce resources while enhancing the quality of engineering and operational follow-up of SIS. Digitalization relies on data exchange in an automated and autonomous way. Mastering interoperability is crucial to fully leverage the benefits of increasing digitalization. Interoperability is a key prerequisite for successful digitalization. It requires standardizing information models and the meaning of the terms being exchanged so they are understood by people, and adding machine-readable codes so they are equally understood by machines.

The submodel Safety Instrumented System (SIS) is one out of four submodels anchored in IEC 61511. The purpose of the submodel is to define and structure relevant properties derived from the safety requirements specification (SRS) allocated to the SIS that realises safety instrumented functions (SIFs). The submodel should be introduced from the stage when information about the functional and safety integrity level requirements becomes available. The submodel is updated with design data during the preliminary and detailed design phase, and operational data is added during operation. Updates must comply with IEC 61511's requirements for management of change. To model the dependencies between the asset Safety Instrumented System and the asset SIS device, the submodel "Hierarchical Structures enabling Bills of Material" can be used.

1.3. Relevant standards for the Submodel template

IEC 61511-1:2016+A1:2017: Functional safety - Safety instrumented systems for the process industry sector - Part 1: Framework, definitions, system, hardware and application programming requirements [7]

NE 192:2025: NAMUR Recommendation 192 - Functional Safety Information Model [8]

APOS Project Report 2023:00108: Guidelines for standardised classification and failure reporting for safety equipment in the petroleum industry [9]

APOS Project Report 2023:00107: Guideline for follow-up of Safety Instrumented Systems (SIS) in the operating phase [10]

1.4. Steps for engineering and operation of Safety Instrumented Systems

If a risk analysis shows that additional safety measures are needed to reach a defined level of safety, a SIF might be required [1] to prevent the process from entering unsafe states. Following the hazard and risk assessment, the first two steps toward developing a SIF are the allocation of protection layers and the preparation of the Safety Requirement Specification for the SIF (dark blue in [\[Figure_1\]](#)) and the Safety Instrumented Systems (SIS) that realize the SIF.

The SIS part (green in [\[Figure_1\]](#)) involves the design and engineering of the SIS and its SIS devices. At this stage, the instrumentation and control engineers specify all SIS devices needed to implement the required safety functions. This includes input devices, final elements, logic solvers, infrastructure devices and electrical components.

The implementation of hardware and software is then validated and commissioned. After commissioning, it becomes the asset in operation (pink in [\[Figure_1\]](#)), representing the Operation and Maintenance phase.

NOTE

figure [Figure_1](#).

Chapter 2. Information set for Submodel “Safety Instrumented System (SIS) for the Process Industries”

2.1. General

While defining Submodels the following three aspects must be considered as suggested in [5]:

Use and economic relevance

Quantifying the cost savings from accidents that should never occur is inherently difficult. However, it is clear that for both emerging sectors like hydrogen and established industries such as chemicals, energy, and pharmaceuticals, accident prevention is essential for public trust and acceptance. Moreover, operational disruptions and unplanned failures can be costly and may compromise the reliability of supply chains, including energy and raw materials.

Value creation potential

The following possible value creation potential that the innovation will have for the different stakeholders are identified:

Operator companies

- Improved safety and productivity – better control of major accidents and increased operational regularity due to enhanced follow-up enabled by interoperability, interoperable electronic industry formats for main safety documentation, all design data and safety requirements readily available in operation, reduced needs for manual updates, improved systems and products based on (real time) feedback from operation
- Increased productivity – reduced purchasing, and operational cost (independent of specific manufacturer solutions, scalability, and flexibility (independent of proprietary solution)

Consultants and engineering companies

- Increased productivity and improved quality of service – interoperable electronic industry formats for main safety documentation, reduced needs for manual updates, document revisions and changes during design
- Increased productivity and sales – reuse of existing models and systems which can be sold internationally and to other industry sectors
- Increased competitive edge – knowledge and competence about I4.0, digital twins (AAS) and associated human centric solutions.
- New markets – better integration of matured concepts from the established industries to emerging sectors

Vendors and manufacturers

- Increased sales – improved systems and products based on (real time) data from operation, new business opportunities/business models
- Increased competitive edge – faster and more cost-effective development of new solutions, competence about I4.0, digital twins (AAS) and associated human centric solutions
- New markets – better integration of matured concepts from the established industries to emerging sectors

Possible functions and interactions

The use cases for this submodel include:

- Digitalizing SRS data for seamless exchange between manufacturers, system integrators and end users, replacing the current approach of sharing SRS data in less suitable formats like pdf, word, and excel. The SRS data are used into various types of software tools for e.g., consistency checks with referenced design documents and risk analyses, compliance checks with technical solutions, and as benchmark for performance monitoring in operation
- Digitalization of SIF design data for seamless exchange between the same listed actors, replacing documents like safety manuals in less suitable formats. SIF design data are stored in several systems, like databases for equipment listing and maintenance management
- Digital twin implementation for SIFs in operation to collect information about demands and proof test for a systematic evaluation of the operational reliability of the SIS as well as for a review of the assumptions in the design phase

Property specification

See chapter 3 "[Information structures and attributes](#)".

2.2. Overview UML model

The information model discussed in this chapter focuses on a Safety Instrumented System (SIS) as a typical asset. This is based on work performed in the APOS project [12]-[14] as well as results from the NAMUR Working Group 1.3 [15].

The SubmodelElements described in section 3 are structured in the following way (see [\[UML_overview_2_2\]](#)):

NOTE	UML diagram for 0 .
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The Submodel template and its instances consists of the following Submodel Element Collection:

- **SRSRequirements** This SMC contains the requirements allocated to the SIS from the SRS requirements in IEC 61511.

Chapter 3. Information structures and attributes

3.1. Properties of the SM SIS

[UML_Submodel_3_1] shows the UML-diagram defining the relevant properties which need to be set.

NOTE UML diagram for 0.

The SubmodelElements for this first level are described as follows. The table convention is explained in Annex A.2.

idShort:	SafetyInstrumentedSystem
Class:	Submodel
semanticId:	https://admin-shell.io/idta/SubmodelTemplate/SafetyInstrumentedSystem/1/0
Parent:	-
Explanation:	The submodel Safety Instrumented System is a collection of properties about the functional safety requirements and information for the lifecycle of a device of a safety instrumented system realising one or more safety instrumented functions.
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] SISName	https://admin-shell.io/idta/cds/SISName/1 gives relevant information about the SIS	[xs:string]	One
[SMC] SRSRequirements	https://admin-shell.io/idta/cds/SRSRequirements/1 SRS requirement(s) according to IEC 61511	[]	One

3.2. Properties of the SMC SRSRequirements

[UML_Submodel_3_2] shows the UML-diagram defining the relevant properties which need to be set.

NOTE UML diagram for SRSRequirements.

The SubmodelElements for this first level are described as follows. The table convention is explained in Annex A.2.

idShort:	SRSRequirements
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/SRSRequirements/1
Parent:	-

Explanation:	SRS requirement(s) according to IEC 61511
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] SurvivabilityRequirement	https://admin-shell.io/idta/cds/SurvivabilityRequirement/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany
[Prop] RequirementForStartingUpAndRestartingSIS	https://admin-shell.io/idta/cds/RequirementForStartingUpAndRestartingSIS/1 SRS requirement(s) according to IEC 61511	[xs:string]	ZeroToMany
[SMC] InterfacingSystemSpecification	https://admin-shell.io/idta/cds/InterfacingSystemSpecification/1 gives relevant information about the SIS	[]	ZeroToMany
[File] PerformanceStandard	https://admin-shell.io/idta/cds/PerformanceStandard/1 gives relevant information about the SIS	[]	ZeroToMany
[File] ApplicationProgramSafetyRequirements	https://admin-shell.io/idta/cds/ApplicationProgramSafetyRequirements/1 SRS requirement(s) according to IEC 61511	[]	ZeroToOne
[File] DALSpec	https://admin-shell.io/idta/cds/DALSpec/1 SRS requirement(s) according to IEC 61511	[]	ZeroToOne
[File] Document	https://admin-shell.io/idta/cds/Document/1 gives information about relevant document(s)	[]	ZeroToMany

Annex A. Explanations on used table formats

1. General

The used tables in this document try to outline information as concise as possible. They do not convey all information on Submodels and SubmodelElements. For this purpose, the definitive definitions are given by a separate file in form of an AASX file of the Submodel template and its elements.

2. Tables on Submodels and SubmodelElements

For clarity and brevity, a set of rules is used for the tables for describing Submodels and SubmodelElements.

- The tables follow in principle the same conventions as in [5].
- The table heads abbreviate 'cardinality' with 'card'.
- The tables often place two informations in different rows of the same table cell. In this case, the first information is marked out by sharp brackets [] from the second information. A special case are the semanticIds, which are marked out by the format: (type)(local)[idType]value.
- The types of SubmodelElements are abbreviated (see Table 1):

Table 1. Abbreviations for SubmodelElements

SME type	SubmodelElement type
Blob	Blob
Cap	Capability
Ent	Entity
Evt	Event
File	File
MLP	MultiLanguageProperty
Opr	Operation
Prop	Property
Range	Range
Ref	ReferenceElement
Rel	RelationshipElement
RelA	AnnotatedRelationshipElement
SMC	SubmodelElementCollection
SME	SubmodelElement type
SML	SubmodelElementList

- If an idShort ends with '___00__', this indicates a suffix of the respective length (here: 2) of decimal digits, in order to make the idShort unique. A different idShort might be chosen, as long as it is unique in the parent's context.
- The Keys of semanticId in the main section feature only idType and value, such as: <https://admin-shell.io/vdi/2770/1/0/DocumentId/Id>. The attribute "type" (typically "ConceptDescription" and "(local)" or

"GlobalReference") need to be set accordingly; see [6].

- If a table does not contain a column with "parent" heading, all represented attributes share the same parent. This parent is denoted in the head of the table.
- Multi-language strings are represented by the text value, followed by '@'-character and the ISO 639 language code: example@EN.
- The [valueType] is only given for Properties.

Annex B. Changes to the submodel template

General

This annex lists the changes from version to version of the Submodel, together with major changes in the overall document.

Changes

This is the first version

Known issues

None

Bibliography

- [1] "Recommendations for implementing the strategic initiative INDUSTRIE 4.0", acatech, April 2013. [Online]. Available: <https://en.acatech.de/publication/recommendations-for-implementing-the-strategic-initiative-industrie-4-0-final-report-of-the-industrie-4-0-working-group/>
- [2] "Implementation Strategy Industrie 4.0: Report on the results of the Industrie 4.0 Platform"; BITKOM e.V. / VDMA e.V., /ZVEI e.V., April 2015. [Online]. Available: <https://www.bitkom.org/sites/main/files/file/import/2016-01-Implementation-Strategy-Industrie40.pdf>
- [3] "The Structure of the Administration Shell: TRILATERAL PERSPECTIVES from France, Italy and Germany", March 2018, [Online]. Available: <https://www.plattform-i40.de/I40/Redaktion/EN/Downloads/Publikation/hm-2018-trilaterale-coop.html>
- [4] "Examples of the Asset Administration Shell for Industrie 4.0 Components – Basic Part"; ZVEI e.V., Whitepaper, April 2017. [Online]. Available: <https://www.zvei.org/en/press-media/publications/examples-of-the-asset-administration-shell-for-industrie-40-components-basic-part>
- [5] "Verwaltungsschale in der Praxis. Wie definiere ich Teilmodelle, beispielhafte Teilmodelle und Interaktion zwischen Verwaltungsschalen (in German)", Version 1.0, April 2019, Plattform Industrie 4.0 in Kooperation mit VDE GMA Fachausschuss 7.20, Federal Ministry for Economic Affairs and Energy (BMWi), Available: <https://www.plattform-i40.de/PI40/Redaktion/DE/Downloads/Publikation/2019-verwaltungsschale-in-der-praxis.html>
- [6] "Details of the Asset Administration Shell; Part 1 - The exchange of information between partners in the value chain of Industrie 4.0 (Version 3.0RC01)", November 2020, [Online]. Available: https://industrialdigitaltwin.org/wp-content/uploads/2021/09/07_details_of_the_asset_administration_shell_part1_v3_en_2020.pdf
- [7] IEC 61511-1:2016+A1:2017: Functional safety - Safety instrumented systems for the process industry sector - Part 1: Framework, definitions, system, hardware and application programming requirements
- [8] NE 192:2025: NAMUR Recommendation 192 - Functional Safety Information Model
- [9] APOS Project Report 2023:00108: Guidelines for standardised classification and failure reporting for safety equipment in the petroleum industry. Available: <https://www.sintef.no/en/publications/publication/2086607/>
- [10] APOS Project Report 2023:00108: Guideline for follow-up of Safety Instrumented Systems (SIS) in the operating phase. Available: <https://www.sintef.no/en/publications/publication/2144020/>
- [11] IEC 61508 Series: Functional Safety of Electrical/Electronic/Programmable Electronic Safety-related Systems (E/E/PE, or E/E/PES)
- [12] PDS Forum: Available: <https://pds-forum.com/>
- [13] APOS 1.0 Web Page: Available: <https://pds-forum.com/about-apos>
- [14] APOS 2.0 Web Page: Available: <https://pds-forum.com/about-apos-2-0>
- [15] NAMUR: Available: <https://www.namur.net/en/index.html>

Additional element tables

The following Submodel Element Collections form part of this Submodel template. Their attribute tables are generated directly from the Submodel template artifact, so they cannot diverge from it.

Properties of SRSRequirements/InterfacingSystemSpecification

idShort:	InterfacingSystemSpecification
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/InterfacingSystemSpecification/1
Parent:	-
Explanation:	gives relevant information about the SIS
Element details:	

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] Interfacing System	https://admin-shell.io/idta/cds/InterfacingSystem/1 gives relevant information about the SIS	[xs:string]	One
[Prop] NetworkType	https://admin-shell.io/idta/SafetyInstrumentedSystem/TypeOfNetwork/1/0 gives relevant information about the SIS	[xs:string]	One
[Prop] TypeOfLinksHardwiredOrSoftware	https://admin-shell.io/idta/SafetyInstrumentedSystem/TypeOfLinks/1/0 gives relevant information about the SIS	[xs:string]	One